

SHAURYA SHARTHAK

Data Scientist

+91 6387864528 | shauryajnvkr@gmail.com | shaurya-sharthak | IsNoobGrammer | HuggingFace

PROFESSIONAL SUMMARY

Data scientist with deep expertise in **large-scale data engineering**, **NLP**, and **statistical analysis**. Engineered **54 datasets (25M+ rows, ~140 GB)** across medical, math, multilingual, and reasoning domains. Built production **hybrid retrieval pipelines** achieving 45% hit-rate improvement. Published first-author research on tokenizer design. Proficient in Python, SQL, Pandas, and end-to-end data pipelines from curation and deduplication to model training.

EXPERIENCE

AI Engineer (Intern)

ResoluteAI Software Inc.

08/2024 – 01/2026

Bangalore, India

- Data-Driven Retrieval:** Designed hybrid pipeline (TF-IDF + BM25 + Neo4j + Redis + vector similarity), improving hit-rate by **45%**.
- Cost Analysis:** Analyzed inference bottlenecks and refactored data flow, achieving **Rs.25.8 Lakhs/year** savings and **30% faster processing**.
- Automated Analytics:** Built Pandas agent for ticket analysis, gaining **30% efficiency** and **20% faster response**.

Data Engineer

TinyCompany-AI

2025

Remote

- Curated **30K-entry Hindi instruct dataset**; developed **HINGLISH-LIMA** for code-mixed NLP; RLHF annotation workflows.

TECHNICAL SKILLS

Data Science: Pandas, NumPy, Scikit-learn, Matplotlib, Statistical Analysis, Feature Engineering, A/B Testing

NLP & ML: HuggingFace (Transformers/Datasets), PyTorch, TensorFlow, LangChain, Tokenizer Design (BPE/SentencePiece)

Data Engineering: MinHash-LSH Deduplication, ETL Pipelines, Quality Filtering, Corpus Balancing, Data Annotation

Languages: Python (Expert), SQL, C++, Bash **Databases:** PostgreSQL, ChromaDB, Pinecone, Neo4j, Redis

Tools: Git/GitHub, Docker, Jupyter, AWS (EC2/S3), REST APIs, Postman

KEY PROJECTS

Large-Scale Dataset Engineering Portfolio | HuggingFace

2024 – 2025

- Engineered **54 datasets (25M+ rows, ~140 GB)** across medical, math, code, reasoning, DPO/RLHF, and multilingual NLP using Mistral/Gemini APIs and manual curation.
- Built **MinHash-LSH dedup pipeline** for large-scale corpus cleaning; designed quality filtering and language-balance metrics for multi-source instruction mixes.
- Created **SymptomsDisease246K** (494K rows) — symptoms-to-disease mapping used by downstream medical chatbots (**53 downloads, 14 likes**).
- Curated **Be-More-Specific-USER** (51K rows) — first-of-its-kind trilingual clarification-seeking dataset (**9 likes**).

Multilingual Data Analysis — Hindi/Hinglish NLP

2024 – 2025

- Designed **6 Hindi/Hinglish CoT datasets** (21K+ rows) covering medical, philosophical, and general reasoning with structured schema.
- Created **Hinglish-Preference-DPO** (9.4K rows, 256 topics) with statistical distribution across 5 domains.
- Built **5M+ deduplicated Hindi corpus**; analyzed language distribution (57.9% EN, 37.1% HI, 5.1% Mixed) across 3.5M-row instruction mix.

QTK-81K Tokenizer — Efficiency Benchmarking | HuggingFace

2025

- Designed 81K BPE tokenizer; benchmarked against Phi-4, DeepSeek-R1, Gemma-2 achieving **4× better Hindi compression**.

Cybersecurity Penetration Testing

2024

- Vulnerability assessments for **3+ MNC web apps** using Burp Suite; secured systems processing **2M+ monthly transactions**.

PUBLICATIONS

Tokenizer Flexibility in LLMs via Heuristic Adaptation and Supertoken Learning

arXiv:2505.09738 | 2025

- Lead Author | 4 citations** — TokenAdapt: zero-shot tokenizer transplantation with **2× perplexity improvement** on Llama-3.2-3B and Qwen2.5-3B.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University

2022 – 2026

B.Tech (Hons.) Computer Science (AI/ML)

Bhilai, India

Jawahar Navodaya Vidyalaya — Dakshana Foundation Scholar (NDST 2018)

2018 – 2021

Senior Secondary (GPA: 3.86/4.0) — Secondary (GPA: 3.6/4.0)

Kurukshetra, India